HARSHAL RAAJ PATNAIK

Senior Software Engineer | Cloud Platforms, Distributed Systems & Technical Lead (Tech × MBA)

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EDUCATION

Indian Institute of Management Kozhikode (IIMK) **2025 – Present**

Executive Post Graduate Programme (EPGP) in General Management – Product Strategy, Finance & Operations *Kochi, India*

SRM Institute of Science and Technology **2017 – 2021**

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.76 / 87.6%) *Chennai, India*

PROFESSIONAL EXPERIENCE

Novamesh Ltd. (Wholly Owned Subsidiary of Tata Communications) **Apr 2024 – Present**

Senior Software Engineer (Cloud Platforms & Distributed Infrastructure) *Chennai, India*

- **Capacity & Infrastructure Telemetry Dashboard:** Architected and engineered end-to-end multi-tenant Capacity Planning & Inventory Dashboard from design to distributed backend services, managing real-time lifecycle records across **Compute** (KVM, VMware ESXi, OLVM hypervisors), **Storage** (NetApp ONTAP, HPE 3PAR/Nimble, Caringo Swarm), and **Network** (public IP pool reports, switch port mappings, VDOM reports, bandwidth utilization telemetry) with interactive graphical heatmaps and trend analytics.
- **Multi-Hypervisor Orchestration & Asynchronous Sagas:** Architected event-driven VM orchestration and capacity tracking engine across **KVM, VMware ESXi, and OLVM**; designed automated compensating sagas with **Flowable BPMN** and decoupled execution via **RabbitMQ (AMQP)** queues to guarantee sub-second portal responsiveness and zero orphaned cloud resources.
- **High-Throughput Batch Processing:** Slashed cloud batch scheduler execution latency by **92.6% (from ~18 minutes down to 80 seconds)** across distributed Cassandra clusters by implementing strict 50-item batch chunking, multi-level Redis caching, and tuning HikariCP connection pools.
- **Fortinet Firewall & Network Security Automation:** Automated enterprise **Fortinet FortiGate Firewalls** via REST APIs for dynamic policy orchestration, VDOM segmentation, and IPsec VPN provisioning; engineered an in-process sliding-window rate limiter that **eliminated 100% of HTTP 429 rate-limiting failures**.
- **API Gateway & Agile Leadership:** Engineered route discovery, wildcard forwarding, and multi-tenant security policies across **Kong API Gateway**; stepped up as **Acting Scrum Master** and technical mentor, driving architecture reviews, code quality, and on-time sprint deliveries.

Tata Communications Ltd. **Jul 2021 – Mar 2024**

Software Development Engineer (Cloud Infrastructure, Storage & Full-Stack) *Chennai, India*

- **Cloud Object Storage & DNS Automation:** Built full-stack onboarding workflow for **Caringo Swarm S3-compatible cloud object storage** and automated cPanel DNS A-record associations using Java and Angular, enabling **100% self-service storage provisioning and automated bucket metering**.
- **Frontend Modernization & Component Architecture:** Spearheaded legacy AngularJS migration to **Angular 17** with reactive component architectures, state predictability, and modular testability, achieving a **35% improvement in client-side rendering speed**.
- **DevOps & CI/CD Pipelines:** Automated continuous integration, regression testing, and release packaging pipelines with **Jenkins CI/CD**, multi-module Maven builds, and JUnit quality gates, significantly cutting manual verification overhead and deployment friction.
- **Network Telemetry & Monitoring:** Engineered switch-port utilization and network capacity telemetry engine integrating Zabbix and Nimsoft APIs, computing cluster-wide compute and datastore allocation to prevent capacity exhaustion; earned **3× Extra Mile Commitment Awards**.
- **Multi-Vendor Load Balancer Orchestration:** Automated enterprise load balancer provisioning and health probe monitoring across **HAProxy, F5 BIG-IP, and VMware Avi** with automated VIP provisioning, traffic steering, and tenant-scoped SSL/TLS offloading.

STRATEGY & TECHNICAL PROJECTS

SthiraX: Distributed CO_2-Mineralization & IoT MRV Platform | *IIM Kozhikode Capstone (BPP)*

- Architected asset-light retrofit platform converting regional concrete plants into certified low-carbon facilities to fulfill India's **April 2026 C&D Waste Rules (mandating 5% to 25% recycled material by 2030 across 20,000+ sqm projects)**; built batch-level **IoT MRV (Measurement, Reporting & Verification) software** generating cryptographically verifiable Digital Product Passports for developer EPR compliance, supported by SaaS and royalty unit economics.

MarginOS: Multi-Tenant Cloud FinOps & Unit Economics Optimization Platform | *Cloud Economics & FinOps*

- Architected distributed FinOps intelligence engine analyzing multi-cloud hypervisor and storage telemetry to calculate real-time gross margins, compute unit economics (cost-per-VM/hour, storage tiers), and surface automated right-sizing recommendations that eliminate up to 28% in infrastructure spend leakages.

Workflow Reliability Lab: Distributed Sagas & Chaos Engineering | *Distributed Systems Platform*

- Engineered developer platform for testing distributed workflow reliability; built synthetic fault injection simulators (network partitions, timeout anomalies) and invariant checkers verifying ACID saga compensations (**Java 25, Spring Boot 4, Angular, Docker**).

QueryLens AI: Sandboxed Enterprise Text-to-SQL & Semantic RAG Engine | *AI & Data Systems*

- Architected modular C4 3-tier Text-to-SQL engine (**Angular 17 → Spring Boot 3 → FastAPI with pgvector RAG**) with schema semantic search, AST validation, and read-only sandbox roles preventing SQL injection and unauthorized data exfiltration.

CloudScale: Predictive Workload Autoscaler & Proactive Telemetry Engine | *Autonomous Cloud Infrastructure*

- Built proactive Kubernetes/VM capacity autoscaling engine integrating Prometheus, Grafana, and time-series ML forecasting models to predict compute resource spikes 15 minutes in advance, mitigating cold-start latencies and reducing over-provisioned idle capacity.

TECHNICAL & LEADERSHIP COMPETENCIES

Languages & Core: Java (8/11/17/25 LTS), Python, TypeScript, JavaScript, SQL, HTML5/CSS3, Bash, Spring Boot 3/4, FastAPI, Angular 17/18, RxJS
Distributed Systems & Cloud: Microservices Architecture, Flowable BPMN, RabbitMQ (AMQP), Event-Driven Architecture, High Availability (HA), Fault Tolerance, KVM, VMware ESXi, OLVM, Docker, Kubernetes
Databases, Caching & Storage: Apache Cassandra (NoSQL), Redis (In-Memory Cache), PostgreSQL (pgvector), MySQL, Caringo Swarm (S3 Object Storage), NetApp ONTAP, HPE 3PAR/Nimble, Hibernate/JPA, HikariCP Tuning
Networking, Security & Firewalls: Fortinet FortiGate Firewalls, VDOMs, Kong API Gateway, IPsec VPN, DNS Automation (cPanel/BIND), Token Bucket Rate Limiting, Reverse Proxies, HAProxy, F5 BIG-IP, VMware Avi
DevOps, SRE & Observability: Jenkins CI/CD, CI/CD Pipelines, Prometheus, Grafana, OpenTelemetry, Distributed Tracing, Zabbix APIs, Nimsoft, JUnit 5, Mockito, SonarQube, Linux/Unix Administration
Product Strategy & MBA Coursework: Product Management, Financial Accounting, Operations Management, Strategy Management, Risk Analysis, Product Analytics, Negotiation & Conflict Management, Entrepreneurship, FinOps, Unit Economics

HONORS, CERTIFICATIONS & EXTRACURRICULAR

- **3× Extra Mile Commitment Award** – Tata Communications Ltd. for critical cloud platform delivery, stability, and zero-defect deployments.
- **Flipkart Grid 2.0 Engineering Challenge** – Team Lead; achieved **Top 150 Nationwide Rank** out of 20,000+ teams (Computer Vision & OCR Track).
- **Architecting with Google Compute Engine** (Google Cloud / Coursera) & **Full Stack Web Development** (Coding Ninjas).
- **Rank 2404 Globally** in Second Round of 5th SOF International Mathematics Olympiad (Science Olympiad Foundation).